Cerberus tools in the VS Code terminal: verification steps

Updated: 2 November 2026 (Cairo) · MacBook (zsh) and Windows (PowerShell)

This guide runs the tool checks from the VS Code terminal. Open the terminal with **Terminal → New Terminal** (on a Mac you can also press `Ctrl+`` or `Cmd+J`).

> **Rules for every step:** run commands from the project root unless a step says **backend**. Test only systems you are allowed to test. No public site is used in these steps.

0. Where data is stored

Item	Stored on disk?	Where	How to remove
Lab results printed in the terminal	No	Terminal window only	Close the terminal
Temporary lab servers and sqlmap files	Briefly	Temporary folders	Removed automatically on normal finish. A forced stop (Ctrl+C twice) can leave files behind.
Python environment	Yes	`backend/.venv`	Keep it. Delete only if you want to rebuild it.
Test cache	Yes	`backend/.pytest_cache`	`rm -rf backend/.pytest_cache` (Mac) or `Remove-Item -Recurse backend\.pytest_cache` (Windows)
Real Metasploit program (optional)	Only if installed	Its install folder, and logs in your user profile (for example `~/.msf4` on a Mac)	Use the uninstaller from Rapid7
Private settings	Yes	`backend/.env`	Keep it private. Do not commit or share it.

The lab scripts do **not** save results to the application database.

1. Set up the Python environment (once)

Mac (zsh):

```
cd backend
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt -r requirements-dev.txt
cd ..
```

Windows (PowerShell):

```
Set-Location backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt -r requirements-dev.txt
Set-Location ..
```

If `backend/.venv` already exists, do not delete it. Use it as it is.

Expected: pip completes without errors.

2. Run the framework unit tests (no database, no external program)

Mac (zsh):

```
cd backend
./.venv/bin/python -m pytest --noconftest -q -k "not persist" tests/unit/test_pentest_tools.py tests/
unit/test_pentest_framework.py tests/unit/test_pentest_safety_regressions.py tests/unit/
test_sqlmap_integration.py tests/unit/test_public_tool_results.py
cd ..
```

Windows (PowerShell):

```
Set-Location backend
.\.venv\Scripts\python.exe -m pytest --noconftest -q -k "not persist" tests/unit/test_pentest_tools.py
tests/unit/test_pentest_framework.py tests/unit/test_pentest_safety_regressions.py tests/unit/
test_sqlmap_integration.py tests/unit/test_public_tool_results.py
Set-Location ..
```

Expected: `228 passed, 2 deselected`. This is the result recorded on Linux; if your count differs, note the first failing test.

3. Run the beginner lab for each tool

These run on a temporary local lab only. Install Nmap and sqlmap first if you want their real checks (see the beginner guide).

Mac (zsh):

```
./backend/.venv/bin/python scripts/live_tests/beginner_tools_lab.py nmap
./backend/.venv/bin/python scripts/live_tests/beginner_tools_lab.py sqlmap
./backend/.venv/bin/python scripts/live_tests/beginner_tools_lab.py metasploit
./backend/.venv/bin/python scripts/live_tests/beginner_tools_lab.py all
```

Windows (PowerShell):

```
.\backend\.venv\Scripts\python.exe scripts\live_tests\beginner_tools_lab.py nmap
.\backend\.venv\Scripts\python.exe scripts\live_tests\beginner_tools_lab.py sqlmap
.\backend\.venv\Scripts\python.exe scripts\live_tests\beginner_tools_lab.py metasploit
.\backend\.venv\Scripts\python.exe scripts\live_tests\beginner_tools_lab.py all
```

Expected:

- `nmap` prints `PASS: real Nmap found the local lab's open port.`
- `sqlmap` prints `PASS: real sqlmap detected the lab's intentional SQL injection in id.`

- `metasploit` prints `SIMULATED` and a `PASS` for the fake response. It does **not** test real Metasploit.
- `all` runs Nmap, then sqlmap, then the simulated Metasploit step, one after another.

A `FAILED` line, a missing-tool message, or an `ERROR` is **not** a pass. Read the message and follow the troubleshooting table in the beginner guide.

4. Test the real Metasploit daemon (optional)

This needs the Metasploit program and `msfrpcd` running. The full steps are in **metasploit steps to check up.pdf**. Do not run it against any public site.

5. Public websites: not run from the terminal

Public captures are refreshed only by the script below. It refuses to run without the confirmation flag, and it only uses the allowed targets:

- `scanme.nmap.org` for Nmap only (about 12 scans per day)
- SekuMart's product `id` parameter for sqlmap only

Mac (zsh):

```
./backend/.venv/bin/python scripts/live_tests/public_tool_results.py nmap --confirm-public-tests
```

Windows (PowerShell):

```
.\backend\.venv\Scripts\python.exe scripts\live_tests\public_tool_results.py nmap --confirm-public-tests
```

Run it only when you need a fresh capture. Do not use these sites for any other tool or purpose. Metasploit has no public target.

6. Clean up after testing

1. Stop any running daemon or server with `Ctrl+C` in its own window.
2. Remove temporary files if a run was stopped by force: check the system temp folder for `sqlmap` or `cerberus` leftovers and delete only those you recognise.
3. Keep `backend/.env` private.

7. Troubleshooting

If you see	Try this
`No such file` for the Python path	Open the project root in VS Code. Check that `backend/.venv` exists.
`Missing Python package`	Run the install step (section 1) with the project's `.venv` Python.
`nmap` or `sqlmap` not found	Reopen the terminal. Set `NMAP_PATH` or `SQLMAP_PATH` for this terminal (see the beginner guide).
`SECRET_KEY` too short	Fix the private `backend/.env`. Do not paste the key into chat.
`Activate.ps1` cannot run (Windows)	Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, then try again.

8. Current limits

- The Mac and Windows commands have not been run on those systems in this project. They were checked on Linux.
- The API, AI agent and interface do not start these tools yet.
- Metasploit has no public test result.